Practical 15: CRUD Operations in a Full-Stack Web Application (CO3)

Code:

```
import sqlite3

conn = sqlite3.connect(":memory:")
cur = conn.cursor()
cur.execute("CREATE TABLE products (id INTEGER PRIMARY KEY, name TEXT, price INTEGER)")

# Create
cur.execute("INSERT INTO products (name, price) VALUES ('Pen', 10)")
cur.execute("INSERT INTO products (name, price) VALUES ('Book', 50)")
conn.commit()
print("After CREATE:")
for row in cur.execute("SELECT * FROM products"):
    print(row)

# Read
print("\nREAD product id=1:")
print(cur.execute("SELECT * FROM products WHERE id=1").fetchone())

# Update
cur.execute("UPDATE products SET price=15 WHERE name='Pen'")
conn.commit()
print("\nAfter UPDATE:")
for row in cur.execute("SELECT * FROM products"):
    print(row)

# Delete
cur.execute("DELETE FROM products WHERE name='Book'")
conn.commit()
print("\nAfter DELETE:")
for row in cur.execute("SELECT * FROM products"):
    print(row)

conn.close()
```

Output:

```
After CREATE:
(1, 'Pen', 10)
(2, 'Book', 50)

READ product id=1:
(1, 'Pen', 10)

After UPDATE:
(1, 'Pen', 15)
(2, 'Book', 50)

After DELETE:
(1, 'Pen', 15)
```